Hasan Bukhari

Machine Learning • Research Engineering • Scientific Computing

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Education

University of Southern Mississippi — **Expected Dec 2027**
B.S. in Computer and Information Sciences; Minor in Biological Sciences — *Hattiesburg, MS*
GPA: 4.00 • President's List • Academic Excellence Scholar

Machine Learning & Research Experience

Independent Machine Learning Researcher — **2026 – Present**
CancerShift and Chromatin Contact Prediction Across Chromosomes — *Independent*

- **CancerShift:** Developed a leakage-controlled multicancer classification pipeline across TCGA, GEO, and CPTAC-GDC cohorts; achieved 99.59% accuracy on a blind microarray cohort and 98.24% on external RNA-seq data.
- Built a 680-gene cross-platform panel and selective-prediction workflow reaching 99.83% accuracy at 85.88% coverage on the evaluated external panel.
- **A17:** Designed and trained a residual MLP using engineered epigenomic and genomic-distance features to predict Hi-C contact structure under chromosome-level holdouts and a locked test set.
- Evaluated generalization and calibration under distribution shift, including failed target-blind calibration experiments; document negative results and methodological limitations for reproducibility.

Data Analyst Intern, Health, Safety & Environment — **Jul 2026 – Aug 2026**
Engro Technical Services — *Karachi, Pakistan*

- Developed Python, FastAPI, Power Query, and analytical pipelines for risk scoring and reporting over 130,000+ operational records spanning incidents, audits, inspections, waivers, and corrective actions.
- Automated daily change monitoring across approximately 1,750 controlled files and built executive Power BI models and DAX measures for operational decision support.

Undergraduate Researcher, Ichthyology Lab — **Spring 2026 – Present**
University of Southern Mississippi — *Hattiesburg, MS*

- Build reproducible Python workflows and chromosome-correspondence analyses for chromosome-scale *Fundulus* genomes, synteny, structural variation, and candidate fusion/fission patterns.

Software Engineering Intern — **Aug 2023 – Feb 2024**
Shahid I.T. Consultants — *Karachi, Pakistan*

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and application maintenance.

Selected Engineering & Research Software

BRACE — 1st Place, VOXO Hatchathon 2026 Bio Track | Python, FastAPI, scikit-learn, SciPy | GitHub — **2026**

- Co-built BRACE (Biological Reliability Assessment across Cohorts and Experiments), a workbench for testing whether biological classifiers generalize across independent cohorts through cross-cohort validation, shift characterization, model evaluation, and reproducible evidence export; shipped 336+ automated tests.

AEGIS — Controlled Document Intelligence Platform | Python, FastAPI, React, TypeScript, Rust/Tauri | GitHub — **2026**

- Engineered a local-first Windows system for deterministic document monitoring, retained-version diffing, SHA-256 integrity verification, governance workflows, analytics, and adversarial regression testing.

BioLitGraph — Biomedical Literature Graph Explorer | Python, Flask, PubMed/PubTator, NetworkX | GitHub — **2026**

- Built a literature-mining system that converts biomedical publications into evidence-linked entity graphs and structured CSV/JSON analysis outputs.

WCRE — Windows Checkpoint/Restore Engine | Rust, Win32, Windows PSS, x64 | GitHub — **2026**

- Developing an experimental Windows x64 checkpoint/restart engine with PSS-backed state capture, persistent checkpoint images, restore planning, and controlled virtual-address reconstruction.

Technical Skills

Programming: Python, Rust, C++, SQL, R; Kotlin, Dart
Machine Learning & Scientific Computing: PyTorch, scikit-learn, pandas, NumPy, SciPy; classification, feature engineering, model evaluation, distribution shift
Research & Data: RNA-seq, microarrays, Hi-C, ATAC-seq, epigenomics; TCGA, GEO, CPTAC, ENCODE
Engineering: Linux/WSL, Git/GitHub, GitHub Actions, FastAPI, Flask, REST APIs, Power BI, DAX, Power Query

Leadership & Honors

Leadership: President & Founder, Student Poets Association and Creative Writing Club, USM.

Selected Honors: 1st Place, VOXO Hatchathon 2026 Bio Track; Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026).